SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005

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CALIBRATION

Valid To: August 31, 2017 Certificate Number: 3467.01

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations[1]:

I. Acoustic

Parameter/Equipment	Range	CMC[2] (±)	Comments
Sound Level Meter	114 dB @ 1 kHz	1.3 dB	In comparison with Dawe (acoustic calibrator)
Acoustic Level Calibrator	94 dB, 104, 114 dB @1 kHz	0.6 dB	CEL Inst. (digital sound level meter)
Tachometer (Non-Contact Type)	(100 to 900) rpm (>900 to 10 000) rpm	0.11 rpm 1.1 rpm	Solex (digital stroboscope & signal generator)

II. Chemical

Parameter/Equipment	Range	CMC[2] (±)	Comments
Conductivity Meter	180 µS/cm 980 µS/cm 1416 µS/cm 1992 µS/cm	11.0 µS/cm 16.0 µS/cm 36.0 µS/cm 26.0 µS/cm	Comparison to standard solution
TDS Meter	90 ppm 490 ppm 996 ppm	5.1 ppm 7.7 ppm 13.0 ppm	

Parameter/Equipment	Range	CMC[2, 4, 5] (±)	Comments
pH Meter	4.02 pH 6.98 pH 9.99 pH	0.026 pH 0.039 pH 0.052 pH	Comparison to standard solution
Kinematic & Dynamic Viscosity Oil Std. 15 to 45°C	<10 mm^2/s (10 to 100) mm^2/s (100 to 1000) mm^2/s (1000 to 10 000) mm^2/s (10 000 to 100 000) mm^2/s	0.21 % + 0.6*R* 0.29 % + 0.6*R* 0.38 % + 0.6*R* 0.60 % + 0.6*R* 0.70 % + 0.6*R*	Comparison to certified viscosity reference standard

III. Fluid Quantities

Parameter/Equipment	Range	CMC[2, 4] (±)	Comments
Gas Flow	(1 to 20) cc/m (>20 to 50) cc/m (>50 to 150) cc/m (>150 to 240) cc/m	(0.5 % + 0.013) cc/m (0.7 % + 0.012) cc/m (0.5 % + 0.06) cc/m (0.4 % + 0.21) cc/m	Airflow calibrator with low flow cell
	(20 to 250) cc/m (>250 to 1500) cc/m (>1500 to 3500) cc/m (>3500 to 5500) cc/m	(0.8 % + 0.3) cc/m (0.25 % + 0.7) cc/m (0.12 % + 0.5) cc/m (0.12 % + 1.0) cc/m	Airflow calibrator with standard flow cell
	(2 to 5) LPM (>5 to 12) LPM (>12to 20) LPM (>20 to 29) LPM	(0.5 % + 0.001) LPM (0.1 % + 0.01) LPM (0.22 % + 0.01) LPM (0.7 % + 0.01) LPM	Airflow calibrator with high flow cell
Volume Measurements (Gravimetric Method)	(1 to 10) µl (>10 to 100) µl (>100 to 500) µl (>0.5 to 5) ml (>5 to 20) ml	0.0045 µl (0.0022 % + 0.0045) µl (0.0008 % + 0.0055) µl (0.2 % + 0.009) µl (0.1 % + 0.01) µl	Mettler Toledo, XP26
	100 µl (>100 to 1000) µl (>1 to 5) ml (>5 to 50) ml (>50 to 200) ml	0.05 µl (0.002 % + 0.02) µl (0.6 % + 0.02) µl (0.1 % + 0.035) µl (0.1 % + 0.1) µl	Mettler Toledo, AG285

Parameter/Equipment	Range	CMC[2, 4] (±)	Comments
Volume Measurements (Gravimetric Method) (cont.)	(2 to 50) ml (>50 to 100) ml (>100 to 500) ml (>500 to 1000) ml (1000 to 4000) ml 50 ml (>50 to 100) ml (>100 to 500) ml (>500 to 1000) ml (>1000 to 2000) ml (>2 to 10) L (>10 to 30) L (4 to 20) L (>20 to 100) L (>100 to 200) L (>200 to 300) L (>300 to 400) L (>400 to 500) L	(0.01 % + 0.008) ml (0.0002 % + 0.008) ml (0.0001 % + 0.008) ml (0.0006 % + 0.008) ml (0.00001 % + 0.008) ml (0.1 % + 0.03) ml (0.05 % + 0.03) ml (0.01 % + 0.05) ml (0.004 % + 0.04) ml (0.002 % + 0.04) ml (1 % + 0.04) ml (0.3 % + 0.04) ml (0.04 % + 0.01) L (0.05 % + 0.01) L (0.004 % + 0.1) L (0.009 % + 0.1) L (0.001 % + 0.1) L (0.007 % + 0.1) L	Mettler Toledo,GG4002-S Mettler Toledo, SG32001 Mettler Toledo, IND 221
Volume [3] (Liquid as Medium)	Up to 2000 ml Up to 500 Liters	0.40 % 0.32 %	Graduated volumetric cylinder, volumetric container
Hydrometers	(0.700 to 0.950) SP.G (>0.950 to 1.070) SP.G (>1.070 to 1.370) SP.G (1.480 to 1.550) SP.G	0.00073 SP.G 0.00032 SP.G 0.00057 SP.G 0.00073 SP.G	In comparison with certified hydrometers

IV. Dimensional

Parameter/Equipment	Range	CMC[2] (±)	Comments
Micrometer (inside /outside)	Up to 100 mm	1.1 µm	Gauge blocks (Mitutoyo M103)

Parameter/Equipment	Range	CMC[2] (±)	Comments
Calipers (All Types)	Up to 300 mm Up to 100 mm >100 mm to 600 mm	12 µm 6.4 µm 13 µm/m + 3.5 µm	Gauge blocks (Mitutoyo M103) Caliper checker
Setting Rods	(25 to 125) mm (150 to 200) mm (225 to 275) mm (300 to 425) mm (2 to 5) in (6 to 8) in (9 to 11) in (12 to 16) in	1.2 µm 2.4 µm 3.4 µm 4.0 µm 41 µin 74 µin 130 µin 170 µin	UMM instrument (Trimos)
Ring Gages	Up to 100 mm (>100 to 400) mm (>400 to 600) mm Up to 4 in (>4 to 8) in (8 to 11) in	1.6 µm 3.6 µm 4.5 µm 62 µin 100 µin 140 µin	UMM instrument (Trimos)
Dial Indicators	Up to 25 mm Up to 1 in	6.1 µm 110 µin	Dial indicator calibrator
Height Gauge/ Vertical Instrument	(5 to 100) mm (>100 to 600) mm (0.5 to 4) in (>4 to 16) in	1.2 µm + 2.0 µm/m 0.3 µm + 10 µm/m 50 µin + 1.5 µin/in 100 µin + 3.0 µin/in	Gauge blocks / setting rods (Mitutoyo)
Caliper Checker / Dimensional Measurement	(20 to 100) mm (>100 to 350) mm (>350 to 600) mm	1.9 µm + 4.0 µm/m 3.0 µm + 2.5 µm/m 0.5 µm + 10 µm/m	Vertical instrument (Trimos)

V. Electrical – DC/Low Frequency

Parameter/Equipment	Range	CMC[2, 6] (±)	Comments
DC Voltage – Generate	(0 to 329.9999) mV (0.33 to 3.299999) V (3.3 to 32.99999) V (33 to 329.9999) V (330 to 1000.000) V	25 µV/V + 1.3µV 14 µV/V + 2 µV 14 µV/V + 30 µV 21 µV/V + 300 µV 22 µV/V + 2.0mV	Fluke 5520A
High Voltage	(1to 5) kV dc (5 to 40) kV pulse	0.1 % +0.0079 kV dc 0.2 % +0.08 kV pulse 0.3 % + 4.4 V pulse	Pipeline Inspection Co. (High voltage meter)
DC Voltage – Measure	(0 to100) mV (0 to 1) V (0 to 10) V (0 to 100) V (0 to 1000) V	50 µV/V + 5.4 µV 40 µV/V + 16 µV 45 µV/V + 55 µV 50 µV/V + 1.3 mV 55 µV/V + 13 mV	Fluke 8845A /8846A
	(0 to 200) mV (0 to 2) V (0 to 20) V (0 to 200) V (0 to 1000) V	10 µV/V + 0.07µV 5.0 µV/V + 1.4 µV 5.0 µV/V + 14 µV 8.0 µV/V + 36 µV 10 µV/V + 0.4 µV	Fluke 8508A
DC Current – Generate	(0 to 329.999) µA (0.33 to 3.29999) mA (3.3 to 32.9999) mA (33 to 329.999) mA (0.33 to 2.99999) A (3.0 to 10) A (>10 to 19) A	180 µA/A + 0.023µA 140 µA/A + 0.02 µA 120 µA/A + 0.3 µA 130 µA/A + 3 µA 0.45 mA/A + 58 µA 0.6 mA/A + 0.5 mA 1.2 mA/A + 0.2 mA	Fluke 5520A
	10 A to 500 A >500 A to 1000 A	(3 mA/A + 0.09) A (3.2 mA/A + 0.09) A	Fluke 5520A / Fluke 5500A (Coil)

Parameter/Equipment	Range	CMC[2, 6] (±)	Comments
DC Current - Measure	(0 to 100) µA (0 to 1) mA (0 to 10) mA (0 to100) mA (0 to 400) mA (0 to 1) A (0 to 3) A (0 to 10) A (0 to 200) µA (0 to 2) mA (0 to 20) mA (0 to 200) mA (0 to 2) A (0 to 20) A	0.65 mA/ A + 0.03 µA 0.61 mA/A + 0.07 µA 0.61 mA/A + 2.3µA 0.61 mA/A + 6.3 µA 0.06 % + 0.11 mA 0.08 % + 0.2 mA 0.12 % + 0.8 mA 0.21 % + 1 mA 42 µA/ A + 0.0023 µA 26 µA/A + 0.003 µA 28 µA/A + 0.03 µA 65 µA/A + 0.82 µA 0.022 % + 0.02 µA 0.048 % + 0.40 mA	Fluke 8845A /8846A Fluke 8508A
Resistance - Generate	(0 to 32.9999) Ω (33 to 329.9999) Ω 330Ω to 3.29999 kΩ (3.3 to 32.9999) kΩ (33 to 329.9999) kΩ 330 kΩ to 3.299999 MΩ (3.3 to 32.99999) MΩ (33 to 100.0000) MΩ	0.029 % 67 µΩ/Ω 64 µΩ/Ω 64 µΩ/Ω 67 µΩ/Ω 100 µΩ/Ω 0.07 % 0.068%	Fluke 5520A
Resistance - Fixed Points	1 kΩ 10 kΩ 100 kΩ 1 MΩ 10 MΩ 100 MΩ 1 GΩ 10 GΩ 100 GΩ	0.00008 kΩ 0.0008 kΩ 0.008 kΩ 0.008 kΩ 0.002 MΩ 0.05 MΩ 0.0005 GΩ 0.01 GΩ 0.2 GΩ	High resistance decade box / Tinsley 4721
Resistance – Fixed Points	0.1 Ω to 1 Ω 1 Ω to 10 Ω 10 Ω to 100 kΩ 100 kΩ to 1 MΩ (1 to 10) MΩ (10 to 100) MΩ	200 µΩ/Ω 30 µΩ/Ω 20 µΩ/Ω 25 µΩ/Ω 100 µΩ/Ω 20 µΩ/Ω	IET HARS SERIES – High accuracy decade resistance substituter

Parameter/Equipment	Range	CMC[2, 6] (±)	Comments
Resistance – Measure	(0 to 10) Ω (0 to 100) Ω (0 to 1) kΩ (0 to 10) kΩ (0 to 100) kΩ (0 to 1) MΩ (0 to 10) MΩ (0 to 100) MΩ	0.011 % + 4.2 mΩ 0.011 % + 6.3 mΩ 0.013 % + 0.012 Ω 0.013 % + 0.12 Ω 0.013 % + 1.32 Ω 0.016 % + 9 Ω 0.055 % + 3 kΩ 0.9 % + 0.04 MΩ	Fluke 8845A / 8846A
	(0 to 2) Ω (0 to 20) Ω (0 to 200) Ω (0 to 2) kΩ (0 to 20) kΩ (0 to 200) kΩ (0 to 2) MΩ (0 to 20) MΩ (0 to 200) MΩ (0 to 2) GΩ	55 μΩ/Ω + 3 μΩ 23 μΩ/Ω + 8 μΩ 13 μΩ/Ω + 0.1 mΩ 11 μΩ/Ω + 0.5 mΩ 10 μΩ/Ω + 0.01 Ω 13 μΩ/Ω + 0.1 Ω 19 μΩ/Ω + 1.55 Ω 30 μΩ/Ω + 0.16 kΩ 0.024 % + 0.008 MΩ 0.2 % + 1 MΩ	Fluke 8508A

Parameter/Range	Frequency	CMC[2, 6] (±)	Comments
AC Voltage – Generate			
(1 to 32.999) mV (33 to 329.999) mV (0.33 to 3.29999) V (3.3 to 32.9999) V (33 to 329.999) V (330 to 1020) V	45 Hz to 10 kHz 45 Hz to 10 kHz 45 Hz to 10 kHz 45 Hz to 10 kHz 45 Hz to 10 kHz 45 Hz to 10 kHz	0.045 % + 1.5 μV 0.025 % + 15 μV 0.02 % + 40 μV 0.02 % + 600 μV 0.024 % + 2.5 mV 0.033 % + 10 mV	Fluke 5520A
AC Voltage – Measure			Fluke 8845A /8846A
(0 to 100) mV (0 to 1) V (0 to 10) V (0 to 100) V (0 to 1000) V	10 Hz to 20 kHz 10 Hz to 20 kHz 10 Hz to 20 kHz 10 Hz to 20 kHz 10 Hz to 20 kHz	0.075 % + 0.048 mV 0.075 % + 0.35 mV 0.075 % + 3.42 mV 0.075 % + 34.2 mV 0.075 % + 260 mV	

Parameter/Range	Frequency	CMC[2, 6] (±)	Comments
AC Voltage – Measure (cont.)			Fluke 8508A
(0 to 200) mV (0 to 2) V (0 to 20) V (0 to 200) V (0 to 1000) V	10 Hz to 20 kHz 10 Hz to 20 kHz 10 Hz to 20 kHz 10 Hz to 20 kHz 10 Hz to 20 kHz	0.022 % + 0.004 mV 0.011 % + 0.03 mV 0.012 % + 0.28 mV 0.012 % + 3.5 mV 0.018 % + 23 mV	
AC Current – Generate			
(29.00 to 329.99) µA (0.33 to 3.2999) mA (3.3 to 32.999) mA (33 to 329.99) mA (0.33 to 1.09999) A (1.1 to 10.9999) A (11 to 20.5)A	45 Hz to 1 kHz 45 Hz to 1 kHz 45 Hz to 1 kHz 45 Hz to 1 kHz 45 Hz to 1 kHz (45 to 100) Hz (45 to 100) Hz	0.18 % + 0.01 µA 0.12 % + 0.1 µA 0.052 % + 2 µA 0.052 % + 20 µA 0.063 % + 0.02 mA 0.08 % + 2.3 mA 0.17 % + 2 mA	Fluke 5520A
10A to 50 A >50A to 100 A >100A to 500 A >500 to 1000 A	(45 to 100) Hz	0.3 % + 0.15 A 0.27 % + 0.2 A 0.32 % + 0.2 A 0.36 % + 0.2 A	Fluke 5520A/ Fluke 5500A Coil
AC Current – Measure			
100 µA 1 mA	10 Hz to 5 kHz 10 Hz to 5 kHz	0.25 % + 0.07 µA 0.16 % + 0.47 µA	Fluke 8846A
(0 to 10) mA (0 to 100) mA (0 to 400) mA (0 to 1) A (0 to 3) A (0 to 10) A	10 Hz to 5 kHz 10 Hz to 5 kHz 10 Hz to 1 kHz 10 Hz to 5 kHz 10 Hz to 5 kHz 10 Hz to 5 kHz	0.2 % + 6.75 µA 0.13 % + 52 µA 0.13 % + 0.46 mA 0.13 % + 0.5 mA 0.25 % + 2 mA 0.2 % + 7 mA	Fluke 8845A /8846A
(0 to 200) µA (0 to 2) mA (0 to 20) mA (0 to 200) mA (0 to 2) A (0 to 20) A	10 Hz to 5 kHz 10 Hz to 5 kHz 10 Hz to 1 kHz 10 Hz to 5 kHz 10 Hz to 5 kHz 10 Hz to 5 kHz	0.038 % + 0.022 µA 0.035 % + 0.26 µA 0.038 % + 0.002 µA 0.038 % + 0.02 µA 0.070 % + 0.30 mA 0.29 % + 0.0023 A	Fluke 8508A

Parameter/Equipment	Range	CMC[2] (±)	Comments
Electrical Calibration of Thermocouples – Generate and Measure			
Type B	(600 to 800) ºC (>800 to 1000) ºC (>1000 to 1550) ºC (>1550 to 1820) ºC	0.55 ºC 0.45 ºC 0.41 ºC 0.44 ºC	Fluke 5520A
Type E	(-250 to -100) ºC (>-100 to -25) ºC (>-25 to 350) ºC (>350 to 650) ºC (>650 to 1000) ºC	0.59 ºC 0.22 ºC 0.20 ºC 0.22 ºC 0.27 ºC	
Type J	(-210 to -100) ºC (>-100 to -30) ºC (>-30 to 150) ºC (>150 to 760) ºC (760 to 1200) ºC	0.33 ºC 0.22 ºC 0.20 ºC 0.23 ºC 0.29 ºC	
Type K	(-200 to -100) ºC (>-100 to -25) ºC (>-25 to 120) ºC (>120 to 1000) ºC (>1000 to 1372) ºC	0.40 ºC 0.24 ºC 0.22 ºC 0.32 ºC 0.48 ºC	
Type N	(-200 to -100) ºC (-100 to -25) ºC (>-25 to 120) ºC (>120 to 410) ºC (>410 to 1300) ºC	0.48 ºC 0.28 ºC 0.25 ºC 0.24 ºC 0.33 ºC	
Type R	(0 to 250) ºC (>250 to 400) ºC (>400 to 1000) ºC (>1000 to 1767) ºC	0.69 ºC 0.46 ºC 0.44 ºC 0.51 ºC	
Type S	(0 to 250) ºC (>250 to 1000) ºC (1000 to 1400) ºC (1400 to 1767) ºC	0.58 ºC 0.47 ºC 0.48 ºC 0.58 ºC	
Type T	(-250 to -150) ºC (>-150 to 0) ºC (0 to 120) ºC (120 to 400) ºC	0.74 ºC 0.30 ºC 0.22 ºC 0.20 ºC	

Parameter/Equipment	Range	CMC[2, 6] (±)	Comments
Electrical Calibration of Thermocouples[3]			
Type J	(-100 to 0) ºC (>0 to 100) ºC (>100 to 1100) ºC	0.32 °C + 0.15 % 0.32 °C + 0.025 % 0.2 °C + 0.03 %	MARTEL MC-1200
Type K	(-190 to 0) ºC (>0 to 1300) ºC	0.42 °C + 0.21 % 0.4 °C + 0.05 %	
Type T	(-200 to 0) ºC (>0 to 390) ºC	0.36 °C + 0.05 % 0.30 °C + 0.01 %	
Type E	(-100 to 0) ºC (>0 to 100) ºC	0.34 °C + 0.3 % 0.3 °C + 0.01 %	
Type R	(>0 to 100) ºC (>100 to 1500) ºC	2.3 °C 1.6 °C + 0.015 %	
Type S	(>0 to 1500) ºC (>100 to 1600) ºC	1.7°C 1.7 °C + 0.032 %	
Type B	(600 to 1800) ºC	1.4 °C + 0.05 %	
Type N	(-190 to 0) ºC (>0 to 1400) ºC	0.54 °C + 0.3 % 0.54 °C + 0.01 %	

Parameter/Equipment	Range	CMC[2, 6] (±)	Comments
Electrical Calibration of Thermocouples[3]			
Type J	(-150 to 0) ºC (>0 to 500) ºC (>500 to 1100) ºC	0.33 °C + 0.18 % 0.34 °C 0.3 °C + 0.016 %	MARTEL MC-1200
Type K	(-190 to 0) ºC (>0 to 1350) ºC	0.42 °C + 0.21 % 0.4 °C + 0.05 %	
Type T	(-200 to 0) ºC (>0 to 380) ºC	0.4 °C + 0.3 % 0.3 °C + 0.01 %	
Type E	(-200 to 0) ºC (>0 to 980) ºC	0.4 °C + 0.3 % 0.3 °C + 0.01 %	
Type R	(>0 to 1700) ºC	1.8 °C	
Type S	(>0 to 1700) ºC	1.8 °C	
Type B	(600 to 1800) ºC	2.0°C	
Type N	(-190 to 0) ºC (>0 to 1280) ºC	0.53 °C + 0.3 % 0.58 °C	
Electrical Calibration of RTD – Generate			Fluke 5520A
Pt 385, 100 Ω	(-200 to -80) ºC (-80 to 0) ºC (>0 to 100) ºC (>100 to 300) ºC (>300 to 400) ºC (>400 to 630) ºC (>630 to 800) ºC	0.061 ºC 0.061 ºC 0.083 ºC 0.11 ºC 0.12 ºC 0.14 ºC 0.27 ºC	
Pt 385, 200 Ω	(-200 to -80) ºC (>-80 to 0) ºC (>0 to 100) ºC (>100 to 260) ºC (>260 to 300) ºC (>300 to 400) ºC (>400 to 600) ºC (>600 to 630) ºC	0.05 ºC 0.05 ºC 0.05 ºC 0.061 ºC 0.14 ºC 0.16 ºC 0.17 ºC 0.19 ºC	

Parameter/Equipment	Range	CMC[2] (±)	Comments
Electrical Calibration of RTD – Generate (cont)			Fluke 5520A
Pt 385, 500 Ω	(-200 to -80) ºC (>-80 to 0) ºC (>0 to 100) ºC (>100 to 260) ºC (>260 to 300) ºC (>300 to 400) ºC (>400 to 600) ºC (>600 to 630) ºC	0.050 ºC 0.061 ºC 0.061 ºC 0.072 ºC 0.094 ºC 0.094 ºC 0.110 ºC 0.130 ºC	
Pt 385, 1000 Ω	(-200 to -80) ºC (>-80 to 0) ºC (0 to 100) ºC (>100 to 260) ºC	0.040 ºC 0.040 ºC 0.050 ºC 0.061 ºC	
Electrical Calibration of RTD- Measure			
Pt 385, 100 Ω	(-180 to 0) ºC (>0 to 550) ºC	0.08 % + 0.1 ºC 0.038 % + 0.1 ºC	Fluke 8846A
Pt 385, 200 Ω	(-180 to 100) ºC (>100 to 500) ºC	0.10 % + 0.08 ºC 0.038 % + 0.1 ºC	
Pt 385, 500 Ω	< 0 ºC (0 to550) ºC	0.10 % + 0.083 ºC 0.038 % + 0.1 ºC	
Pt 385, 1000 Ω	(-180 to 0) ºC (>-100 to 550) ºC	0.11 % + 0.072 ºC 0.35 % + 0.1 ºC	
Pt 385, 100 Ω	(-190 to 0) ºC (>0 to 780) ºC	0.26 ºC + 0.012 % 0.18 ºC + 0.1 %	Martel MC-1200
Pt 385, 200 Ω	(-190 to 0) ºC (>0 to 620) ºC	0.93°C 0.93 ºC + 0.06 %	
Pt 385, 500 Ω	(-190 to 0) ºC (>0 to 620) ºC	0.48 ºC 0.48 ºC + 0.06 %	
Pt 385, 1000 Ω	(-190 to 0) ºC (>0 to 620)°C	0.25 ºC 0.30°C + 0.04%	

VI. Mechanical

Parameter/Equipment	Range	CMC[2, 4] (±)	Comments
Pressure – Measure	(150 to 1000) mmH_20 (>1000 to 10 000) mmH_20 (1 to 100) psi (10 to 10 000) psi (1 to 700 bar) (10 to 10000) psi (>10 000 to 20 000) psi (>20 000 to 36 000) psi	0.046 % 0.023 % 0.046 % 0.032 % 0.026 % 0.037 % 0.048 %	Dead weight tester, Budenberg 551 Dead weight tester, Budenberg 350 Dead-weight tester, Budenberg 580DX Dead-weight tester, Budenberg 580EHX
Pressure Valves / Hydrostatic pressure Testing[3] Pressure Gauge / Pressure Measurements	(20 to 6000) psi Up to 5000 psi	0.10 % of F.S. (0.12 % rdg.) psi	Test gauges Pressure calibrator / digital gauges (Crystal XP2i)
Torque	Up to 20.0 N·m (>20 to 50) N·m (>50 to 200) N·m (>200 to 400) N·m (>400 to 1000) N·m (>1000 to 1200) N·m (>1200 to 1800) N·m (>1800 to 2400) N·m (>2400 to 3000) N·m	0.26 % of rdg 0.37 % of rdg 0.24 % of rdg 0.21 % of rdg 0.18 % of rdg 0.13 % of rdg 0.12 % of rdg 0.12 % of rdg 0.12 % of rdg	Torque transducers

Parameter/Equipment	Range	CMC[2, 4, 5] (±)	Comments
Force-Compression & Tension	(10 to 100) kN (>100 to 500) kN	0.93 % 0.58 %	Load cell , dynamometer
Universal Testing Machine[3] (Tension and Compression)	(100 to 1000) kN	1.3 kN	Field calibration using a universal strain gauge & load cell
Compression	Up to 10 KN (>10 to 30) KN (>30 to 50) KN (>50 to 300) KN (>300 to 1000) KN (>1000 to 3000) KN	±0.068% ±0.043% ±0.026% ±0.070% ±0.031% ±0.011%	Load Cell
Tension / Weights and Test Weights - Measurement	0 to 1000 kgf 0 to 2500 lbf 0 to 10 000 lbf 0 to 50 000 kgf	0.02 % + 0.06 kgf + 0.6*R* 0.07 % + 0.28 lbf + 0.6*R* 0.08 % + 1.8 lbf + 0.6*R* 0.08 % of F.S + 0.6*R*	S-load cell Dillon dynamometer
Set of Weights Class E2 ,F1, F2, M1 M2, M3	(1 to 10) mg (>10 to 100) mg (>100 to 500) mg (>500 to 5000) mg (>5 to 20) g	4.5 µg (2.2 % + 4.5) µg (0.8 % + 5.5) µg (0.2 % + 9) µg (0.1 % + 0.01) mg	Digital comparator Mettler Toledo, XP26 / Class E2
	(>200 to 1000) g	1.6 mg	Ohaus EX1103M, class E2
Test Weights	Up to10 g (>10 to 50) g (>50 to 200) g	0.062 mg (0.11 % + 0.03) mg (0.05 % + 0.1) mg	Mettler Toledo, AG 285

Parameter/Equipment	Range	CMC[2, 4] (±)	Comments
Set of Weights Class E2, F1, F2 Class F1, F2 Class F2 Class M1, M2, M3	 10 kg 20 kg 5 kg 1 kg , 2 kg 500 g 1 kg 2 kg 5 kg 10 kg 20 kg 500kg	 15 mg 28 mg 14 mg 12 mg 10 mg 18 mg 10 mg 0.10 g 0.15 g 0.25 g 0.032 kg	 Mettler Toldeo, XP64003L Mettler Toledo, GG40002-S Mettler Toledo, SG32001 Mettler Toledo, IND221
Scales and Balances[3] Accuracy Class I Accuracy Class II Accuracy Class III and Class IIII Weighing and Plant Scales (Build-up)	 (1 to 10) mg (>10 to 100) mg (>100 to 500) mg (0.5 to 5) g (>5 to 50) g (>50 to 200) g (>200 to 500) g (20 to 100) mg (>100 to 500) mg (>0.5 to 5) g (>5 to 50) g (>50 to 300) g (>0.3 to 2) kg (>2 to 10) kg (>10 to 30) kg 500 g to 2 kg (>2 to 20) kg (>20 to 500) kg (>500 to 2000) kg (>500 to 5000) kg	 0.002 mg 0.003 % + 0.002 mg 0.002 % + 0.002 mg 0.12 % + 0.01 mg 0.02 % + 0.02 mg 0.02 % + 0.05 mg 0.03 % + 0.10 mg 0.003 % + 0.002 mg 0.002 % + 0.002 mg 0.12 % + 0.01 mg 0.02 % + 0.02 mg 0.02 % + 0.05 mg 0.1 % + 0.012 g 0.002 % + 0.012 g 0.07 % + 0.012 g 0.1 % + 0.01 g 0.1 % + 0.2 g 1.3 % + 0.2 g 1.1 % + 1 g 0.012 % kg	Class E2 Class E2, F1, F2 Class M1 Class M1 Class M1 & M2
Cable Ladder / Tray Deflection Load Test	Up to 200 mm	0.016 mm	Digital depth gauge

Parameter/Equipment	Range	CMC[2, 4, 5] (±)	Comments
Cable Ladder / Tray Destruction Load Test	Up to 500 kg >500 kg	0.013 kg 1.3 % + 0.2	Field standard weight/digital platform balance
Nuclear Moisture / Density Gauge	Up to 200 pcf Up to 3200 kg/m^3	0.05 pcf + 0.6*R* 1.2 kg/m^3 + 0.6*R*	NDG Validator

VII. Thermodynamics

Parameter/Equipment	Range	CMC[2] (±)	Comments
Temperature – Generate / Measure	(-15 to 200) ºC Ambient to 300 ºC (>300 to 600) ºC	0.11 ºC 0.13 ºC 0.16 ºC	Calibration bath Dry-well calibrator
Temperature - Measure	(-80 to 200) °C (>200 to 600) °C 0 to 60 °C	0.052 °C 0.090 °C 0.13 °C	Standard platinum resistance thermometer/ reference multimeter
Humidity - Measure	(20.0 to 80.0) % RH	2.7 % R.H	High accuracy thermo-Hygrometer
Ovens[3] (Direct Verification) Chiller and Incubator[3] (Direct Verification)	Up to 200 °C -15°C to 100 °C (>100 to 600) °C	0.77 °C 0.50 °C (0.2 %+0.3)°C	Multifunction calibrator (probe) Multifunction calibrator Thermocouple probe
Humidity Measure	(20.0 to 80.0) % RH	0.75 % RH	High accuracy thermo-hygrometer / humidity chamber
Humidity Chamber	(20.0 to 70.0) % RH	0.53 % RH	High accuracy thermo-hygrometer

VIII. Time & Frequency

Parameter/Equipment	Range	CMC[2] (±)	Comments
Frequency – Measure	Up to 10 Hz (>10 to 100) Hz >100 Hz to 1 kHz (>1 to 10) kHz (>10 to 100) kHz >100 kHz to 1 MHz (>1 to 10) MHz (>10 to 100) MHz	0.008 mHz 0.20 mHz 0.0004 Hz 0.004 Hz 0.04 Hz 0.0004 kHz 0.004 kHz 0.008 kHz	HP 5315A universal counter
Frequency Measuring Equipment			Fluke 5820A
Voltage	10 Hz to 10 kHz	0.38 parts in 10^6 s/s	
Edge	1 kHz to 100 kHz >100 kHz to 10 MHz	0.39 parts in 10^6 s/s	
Marker	2 nsec to 10 nsec >10 nsec to 500 msec >0.52 sec to 1 sec >1 sec to 5 sec	0.39 parts in 10^6 s/s 2.9 parts in 10^6 s/s 8 parts in 10^6 + 0.66 µs 8 parts in 10^6 + 0.13 ms	

[1] This laboratory offers commercial calibration service and field calibration service

[2] Calibration and Measurement Capability (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. Calibration and Measurement Capabilities represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of k = 2. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

[3] Field calibration service is available for this calibration and this laboratory meets A2LA *R104 – General Requirements: Accreditation of Field Testing and Field Calibration Laboratories* for these calibrations. Please note the actual measurement uncertainties achievable on a customer's site can normally be expected to be larger than the CMC found on the A2LA Scope. Allowance must be made for aspects such as the environment at the place of calibration and for other possible adverse effects such as those caused by transportation of the calibration equipment. The usual allowance for the actual uncertainty introduced by the item being calibrated, (e.g. resolution) must also be considered and this, on its own, could result in the actual measurement uncertainty achievable on a customer's site being larger than the CMC.

[4] In statement of CMC, percentages are percentage of reading, unless otherwise indicated.

[5] In the statement of CMC, *R* is the resolution of the unit under test.

[6] The stated measured values are determined using the indicated instrument (see Comments). This capability is suitable for the calibration of the devices intended to measure of generated the measured value in the ranges indicated. CMC's are expressed as either a specific value that covers the full range or as a fraction/percentage of the reading plus a fixed floor specification.

Accredited Laboratory

A2LA has accredited

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Al Khobar 31952, SAUDI ARABIA

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets R205 – Specific Requirements: Calibration Laboratory Accreditation Program. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (*refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009*).

Presented this 1st day of September 2015.

President and CEO
For the Accreditation Council
Certificate Number 3467.01
Valid to August 31, 2017

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.